

Rayen Braiek

Software Engineering Student — Backend Systems • AI Integration • Secure Architecture

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EDUCATION

Higher Institute of Computer Science and Mathematics of Monastir Monastir, Tunisia
Software Engineering Degree Sep 2022 – Sep 2027
– Relevant coursework: Machine Learning, Computer Vision, Software Architecture, Databases, Algorithms

EXPERIENCE

Software & AI Engineer Sep 2025 – present
PredictoEnergy Remote
– Contributed to a production SaaS platform for energy analytics and operational decision support.
– Implemented role-based authorization (RBAC) using **Spring Security**, leveraging **Aspect-Oriented Programming (AOP)** to manage cross-cutting security concerns and enforce structured access control.
– Integrated **JSR-354 Money API** to ensure precise and consistent financial calculations across services.
– Architected secure **file storage** using UUID identifiers, metadata isolation, and header sanitization to mitigate injection risks.
– Deployed a **production-grade forecasting and anomaly detection system** leveraging environmental data (temperature and humidity), achieving **92% accuracy** and enabling early identification of overheating and abnormal machine behavior.

Computer Vision Intern – Medical Imaging June 2025 – August 2025
Research Center, Faculty of Medicine of Monastir Monastir, Tunisia
– Developed multimodal **image registration** pipeline capable of aligning medical images, improving disease progression visualization and enhancing anomaly detection and segmentation performance.
– Built and trained deep learning models using **PyTorch** and deployed backend services with **Flask**.

PROJECTS

Flight Delay & Weather Data Engineering Pipeline April 2026
Python, Pandas, PostgreSQL, Apache Kafka, Apache Airflow, Flask, Streamlit
– Built an end-to-end **hybrid ELT pipeline** combining **batch processing** for flight data ingestion with **Kafka-based weather streaming** into PostgreSQL.
– Designed a **Medallion architecture** (Bronze/Silver/Gold) with data cleaning, dual weather joins, and feature engineering for delay, recovery, and seasonality analysis.
– Orchestrated workflows using **Apache Airflow** and a **Flask-based orchestrator**, enabling modular and parallel data processing.
– Developed **analytics KPIs** (weather impact, airport/airline delay rates, recovery rate, seasonal trends) and a **rule-based delay prediction module**, delivered via an interactive **Streamlit dashboard**.

Multimodal AI Retrieval System February 2025
Python, ChromaDB, BGE, CLIP, Whisper, Tesseract OCR, Qwen 2.5, Ollama
– Built a **multimodal AI retrieval system** using vector embeddings to enable semantic search across text, images, and audio.
– Designed **batch ingestion pipelines** for multimodal data, enabling scalable and efficient indexing.
– Integrated a **local LLM** (Qwen 2.5 via Ollama) for contextual question-answering with session management.

CERTIFICATIONS

Building LLM Applications With Prompt Engineering NVIDIA
Supervised Machine Learning: Regression and Classification Stanford — Online
Advanced Learning Algorithms Stanford — Online

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript
Frameworks: Spring Boot, Spring Security, React, Next.js, Flask, PyTorch
AI/ML: OCR, Embeddings, Vector DBs, Deep Learning, Time Series, Ollama
Tools: Git, REST APIs, IntelliJ, VS Code, Agile/Scrum, GitHub Actions, Docker